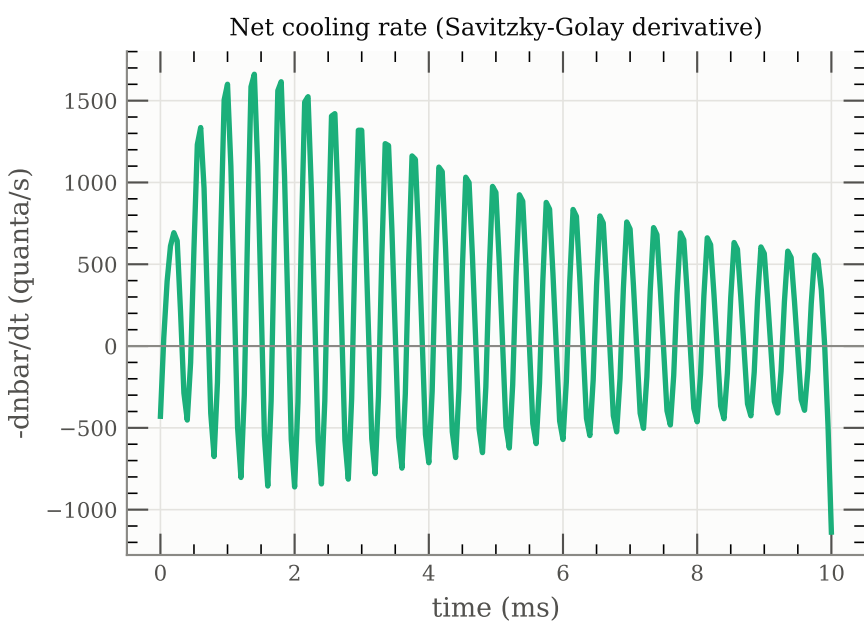
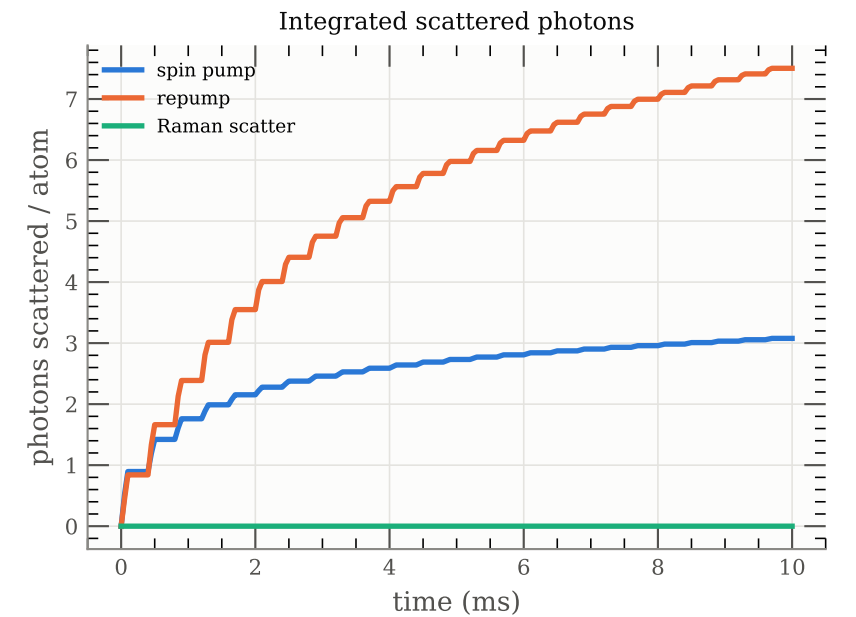
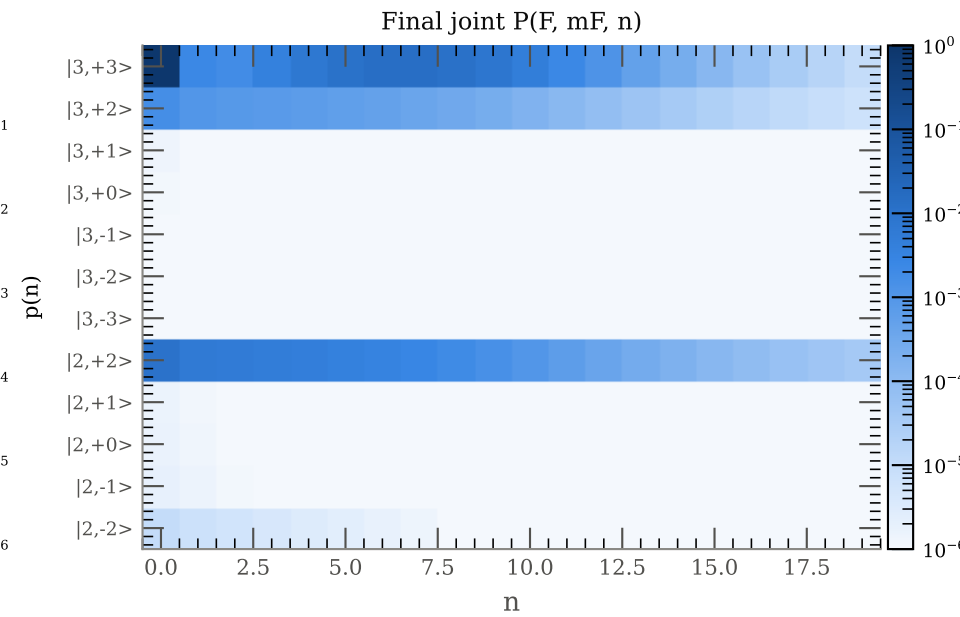
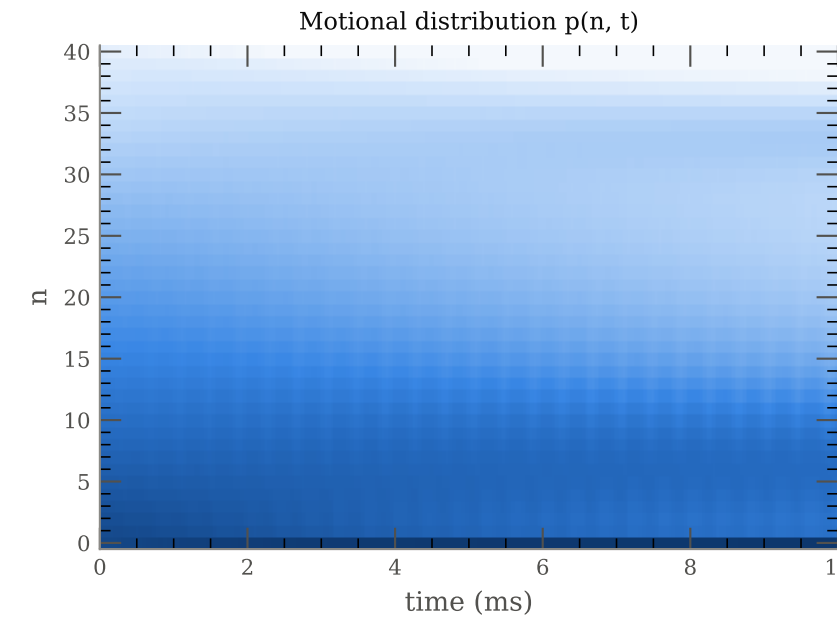
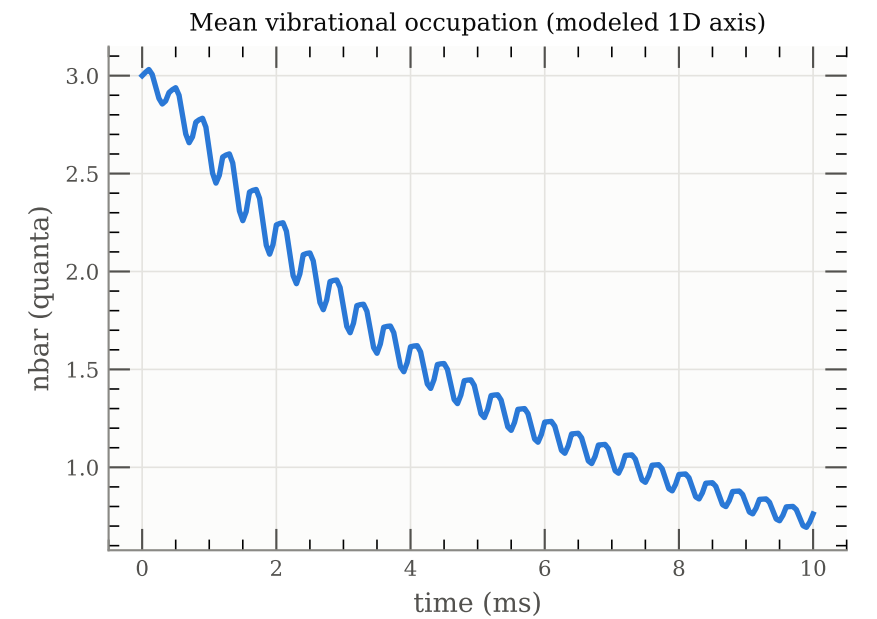
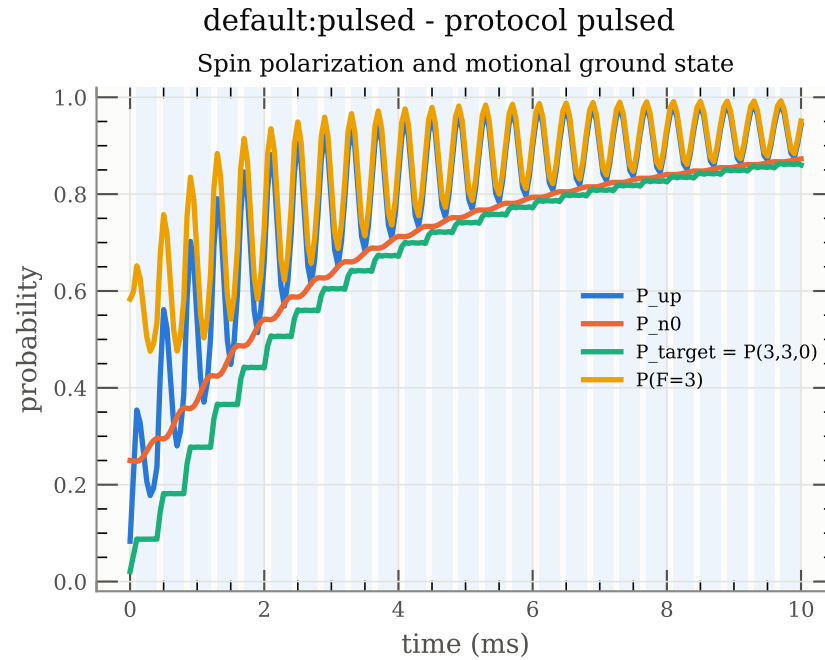
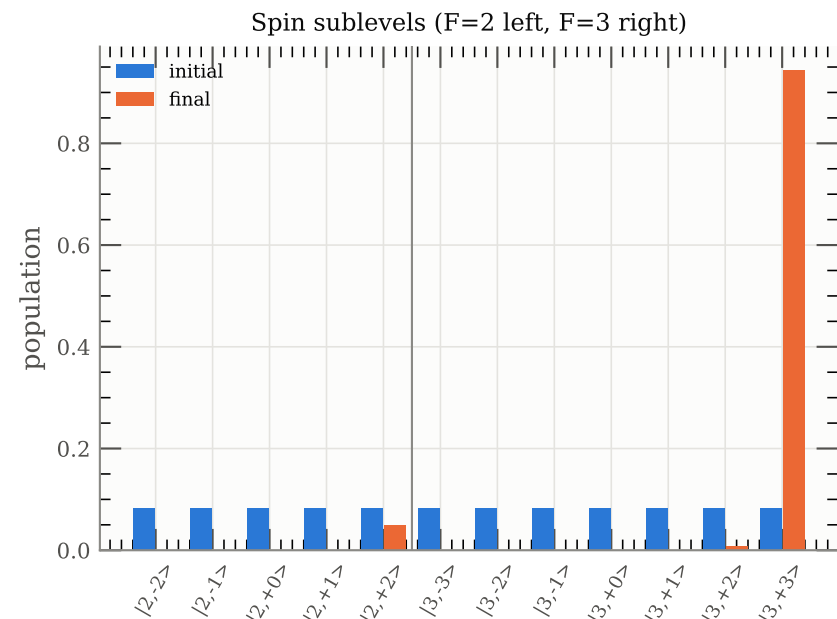




MODEL VALIDITY: WARNING
 $P_{up}=0.9430$ $P_{n0}=0.8727$ $P_{target}=0.8603$ ($P_{up}P_{n0}=0.8229$)
 $nbar\ 3 \rightarrow 0.7617$ $T_{equiv}=4.01\ \mu K$ [PROXY ONLY: final $p(n)$ non-thermal (TVD from thermal 0.38)]
 photons/atom: pump 3.08, repump 7.5, Raman 0
 target $|3,3\rangle$ scattering rate 0 /s; quanta removed per photon 0.21150898271318727
 $t(P_{up}\geq 0.9) = 2.49\ ms$; $t(P_{target}\geq 0.5) = 2.08\ ms$
 conservation err 6e-15; min eig 2.5e-22; boundary pop 5.9e-06; overflow 8e-06
 solver coherent, wall 16.1 s

[OK] magnetic field: weak-field Zeeman OK (quadratic error on ν_{ud} 89.8 Hz, excited quadratic shift 0.0049 Gamma)
 [OK] ground coherences: adjacent-mF Zeeman splitting / max pumping rate = 34.1 (neglect of pump-induced Zeeman coherences needs $\gg 1$)
 [OK] motional secular: max optical scattering rate / $\omega_{at} = 0.0978$ (secular recoil jumps need $\ll 1$)
 [OK] Raman isolation: only $|3,3\rangle \rightarrow |2,2\rangle$ Raman pair modeled; nearest unmodeled channel $|3,+2\rangle \leftrightarrow |2,+2\rangle$ (sideband +2) is 163.4 kHz from the drive = 32.7 x carrier Rabi (assumes comparable coupling)
 [WARNING] Raman scattering: IDEALIZED ASSUMPTION: residual Raman photon scattering rate = 0 (unknown)
 [OK] Raman polarization: calibrated mode: Raman polarization imperfections enter only via the calibrated extra decay, shift, and scattering inputs
 [OK] Lamb-Dicke: $\eta_R = 0.3309$, $\eta_R^2(2\ nbar+1) = 0.766$; exact displacement elements used (no LD expansion)
 [OK] harmonic approximation: trap depth not given: anharmonicity/finite depth not assessed
 [OK] conservation: max $|trace + overflow - 1| = 6e-15$
 [OK] positivity: most negative eigenvalue/population = 2.55e-22
 [OK] motional boundary: max population in $n \geq n_{max}-1$: 5.87e-06
 [OK] numerical overflow: probability pushed above n_{max} (numerical, not atom loss) = 7.96e-06
 [OK] weak excitation: max excited-state fraction for occupied ground states = 0.00113
 [OK] initial state: thermal $nbar=3$; discarded tail above n_{max} before normalization = 7.54e-06